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Helpful or harmful? A curvilinear perspective on AI agent anthropomorphism in service failure tolerance

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ABSTRACT

The anthropomorphic design of AI agents often evokes mixed responses, yet it remains a common strategy to mitigate the negative impact of service failures. This study examines how AI agent anthropomorphism influences customer tolerance for service failures. Through three experiments involving 526 participants in video- and text-based scenarios, the findings indicate a curvilinear relationship between anthropomorphism and failure tolerance, following an inverted U-shape. The study investigates the underlying mechanism driving this effect, with likability serving as a mediator in the curvilinear relationship. Additionally, this study identifies boundary conditions of the effects, where failure severity (1) attenuates the curve relationship between anthropomorphism and failure tolerance and (2) weakens the indirect effect of likability. These findings contribute new factors to cognitive appraisal theory (i.e., likability to personal factor and failure severity to situational factor) and expand uncanny valley theory from emotional to behavioral responses (i.e., tolerance behavior). Practically, this study provides valuable insights for the utilization and deployment of AI agents in customer service.

1. Introduction

AI agents have gained attention across academia, industry, and the public for their autonomous environmental perception and goal-oriented actions, promoting extensive application potential and socio-economic impact (Han et al., 2023; Shehawy et al., 2025). According to the McKinsey (2024) report, 55 % of organizations integrate AI into at least one business unit or function, an increase from 50 % in 2022 and 20 % in 2017. Although AI agents, such as service robots and virtual assistants, have been widely adopted and impactful in various sectors, their implementation in the service industry has been slower and more cautious. The 2024 AI Index Report published by the Stanford Institute for Human-Centered Artificial Intelligence points out that 52 % of customers felt more concern than excitement about AI in 2023, compared to 38 % in 2022 (Maslej et al., 2024). This is because AI agents often struggle with the complex and varied social interactions in the retail and service industry, leading to instances of service failure (Arikan et al., 2023; Choi et al., 2021; Park et al., 2023). Such failure not only harms customer experience but may also impact a company's reputation and interests (Cameron et al., 2021; Mende et al., 2019; Park et al., 2023). Therefore, it is critical for the service industry to minimize the negative effects of service failures by enhancing customer tolerance.

Anthropomorphism is defined as the tendency to attribute human-like characteristics, motivations, intentions, or emotions to non-human entities (Bartneck et al., 2009; Epley et al., 2007). In recent years, anthropomorphism has gained increasing attention in management and marketing research (e.g., Akdim et al., 2023; Mehmood et al., 2024) and has been found to play a crucial role in shaping customer attitudes (Shehawy et al., 2025). However, the effect of AI agent anthropomorphism on customer failure tolerance remains debated (Akdim et al., 2023; Blut et al., 2021; Xie and Lei, 2022; Yang et al., 2024). Some studies indicate that customers exhibit higher failure tolerance toward highly anthropomorphized AI agents (e.g., Arikan et al., 2023; Zheng et al., 2025), whereas others propose that less anthropomorphized AI agents are more likely to increase customer failure tolerance (e.g., Cheng, 2023; Cui and Zhong, 2023; Zhang et al., 2023). Therefore, to address the mixed findings on whether AI agent anthropomorphism increases or decreases failure tolerance, the subject of this study is to further explore how AI agent anthropomorphism influences customer failure tolerance.

In particular, this study attempts to investigate the curvilinear effect of AI agent anthropomorphism on service failure tolerance from

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perspectives of the cognitive appraisal theory and the uncanny valley theory. The cognitive appraisal theory tackles how individuals evaluate environmental events to further affect emotional response and following behavioral response (Lazarus and Folkman, 1984), which helps explain how AI agent anthropomorphism influences customers' emotional responses and service failure tolerance to AI agents during service failures. Meanwhile, the uncanny valley theory offers a framework for understanding how the increasing anthropomorphism of AI agents can lead to negative emotional reactions when AI agents' appearance or behavior becomes too human-like, yet still fails to perfectly resemble real humans.

According to the cognitive appraisal theory, both internal personal factors and external situational factors influence the cognitive appraisal process (Lazarus and Folkman, 1984). Likability, a core element in service encounters (Li et al., 2019), is an internal emotional response elicited by anthropomorphized AI agents (Mathur and Reichling, 2016). It shapes customer behavior in service encounters (e.g., Jin and Hwang, 2024; Nguyen et al., 2013). While previous research has highlighted cognitive factors such as performance expectancy (e.g., Cui and Zhong, 2023; Zhang et al., 2023) and failure attribution (e.g., Arikan et al., 2023; Cheng, 2023) in the anthropomorphism-tolerance relationship, the role of emotional factors remains underexplored.

Failure severity, as an external situational factor, is critical in service failure encounters, because it affects customer emotional states and behavioral responses (Hess, 2008; Wang et al., 2011; Weun et al., 2004). Specifically, more severe failure tends to evoke stronger negative emotions and negative behavioral responses (Wang et al., 2011; Weun et al., 2004), such as decreased failure tolerance (Park et al., 2023). However, existing research on the relationship between AI agent anthropomorphism and failure tolerance has largely overlooked failure severity, particularly its interaction with anthropomorphism and likability in shaping failure tolerance under different conditions.

Therefore, to bridge these research gaps, this study examines how AI agent anthropomorphism affects customer failure tolerance, and the roles of likability and failure severity in this relationship via three video- and text-based experiments. Specifically, we target to address the following research questions.

RQ1. How does AI agent anthropomorphism impact both likability and failure tolerance?
RQ2. How does likability mediate the curvilinear relationship between AI agent anthropomorphism and failure tolerance?
RQ3. How does failure severity moderate the curvilinear relationship between AI agent anthropomorphism, likability, and failure tolerance?

This study has important theoretical and practical contributions. First, this study reveals two curvilinear relationships showing that AI agent anthropomorphism has an effect on failure tolerance and likability following an inverted U-shape, integrating and extending the previously mixed findings (e.g., Arikan et al., 2023; Cheng, 2023; Cui and Zhong, 2023). Second, this study enriches the cognitive appraisal theory (Lazarus and Folkman, 1984) and the uncanny valley theory (Mori, 1970) in the context of AI agent anthropomorphism by identifying likability as an internal emotional mechanism and failure severity as a boundary condition in the anthropomorphism–tolerance relationship. Finally, this study also offers managerial implications for understanding the design and deployment of AI agents in customer service, emphasizing AI agent design should balance anthropomorphism and emotional value, while avoiding tasks prone to severe service failures.

2. Theoretical background and hypothesis development

2.1. Cognitive appraisal theory

We draw on cognitive appraisal theory to explain how AI agent anthropomorphism influences failure tolerance through likability and under varying levels of failure severity. This theory suggests that individuals' emotional responses to environmental stimuli arise not automatically but through a cognitive appraisal process (Lazarus and Folkman, 1984; Smith and Ellsworth, 1985). In this process, individuals first assess whether an event poses a threat to their well-being, viewing it as a potential source of stress. They then select coping strategies, which may involve problem-solving or regulating their emotional response. This appraisal process is influenced by both personal and situational factors (Lazarus and Folkman, 1984), shaping how each person experiences and manages stress.

In the context of service failures, customers often perceive failures as stressful events that threaten their well-being, triggering negative emotional reactions. To cope with these negative emotions, customers may employ failure tolerance as a strategy. Failure tolerance refers to the degree to which customers are willing to endure substandard service quality before considering switching service providers or discontinuing the service (Cui and Zhong, 2023; Zhang et al., 2023). Failure tolerance is a common mechanism for managing the stress associated with failed service encounters (Lv et al., 2021).

2.2. Uncanny valley theory

The uncanny valley theory provides another important perspective for understanding the impact of AI agent anthropomorphism on failure tolerance. Proposed by Mori (1970), this theory explains the non-linear emotional responses humans exhibit toward anthropomorphic objects, such as robots and AI agents. Specifically, it posits that as an AI agent's appearance and behavior become more human-like, positive emotions such as affinity or trust initially increase. However, once anthropomorphism reaches a certain threshold, subtle imperfections in human resemblance can provoke strong negative reactions, such as eeriness, fear, or even disgust (Söderlund, 2022), resulting in what Mori (1970) termed the "uncanny valley." This effect implies a distinct human sensitivity to"imperfect human likeness." Beyond this valley, as anthropomorphic qualities become nearly indistinguishable from those of real humans, positive emotional responses rise again and stabilize (Mori, 1970).

In this study, the uncanny valley theory helps explain how AI agent anthropomorphism influences customer emotions during service failure encounters. Specifically, when AI agents become highly human-like, the initial positive emotions arising from perceived familiarity and identity shift toward negative emotions such as discomfort and identity threat (Blut et al., 2021; Mende et al., 2019; Yang et al., 2024). This shift can undermine customer failure tolerance. Hence, the uncanny valley theory provides a theoretical foundation for explaining why high anthropomorphism may backfire in service failures, leading to lower failure tolerance.

Additionally, the uncanny valley theory provides a complementary perspective to the cognitive appraisal theory in understanding customer reactions to AI agent service failures. While the cognitive appraisal theory explains how individuals assess and interpret the failure event, the uncanny valley theory specifies the role of AI agent anthropomorphism in shaping these appraisals. By integrating these two theories, this study develops a comprehensive framework to explain the nuanced relationship between AI agent anthropomorphism, likability, failure severity, and failure tolerance.

2.3. The inverted U-shaped of AI agent anthropomorphism and failure tolerance

Managers often employ anthropomorphized AI agents to enhance customer interaction by endowing them with human-like traits, such as motivations, intentions, and emotions (Epley et al., 2007). The literature in psychology and marketing emphasizes the importance of anthropomorphism in understanding how customers perceive inanimate objects.

Studies suggest that anthropomorphism improves the acceptance of AI agents by facilitating cognitive, emotional, and associative connections (Xie and Lei, 2022).

Existing research presents mixed findings on the effects of anthropomorphism. A prevalent view is that anthropomorphized AI agents are more easily accepted, as humans have a natural tendency to relate to entities with human characteristics. For example, Mehmood et al. (2024) find that anthropomorphic characteristics of AI service robots have a positive influence on customer engagement. Mende et al. (2019) suggest that customers can empathize with human-like features and emotions in AI agents. This viewpoint is also supported by Choi et al. (2021), who find that anthropomorphic AI agents are more likely to elicit warm feelings in customers. This tendency enables anthropomorphic AI agents to inspire positive perceptions and emotional connections and further helps fulfill human needs for belonging and other emotional connections (Liao and Huang, 2024; Yang et al., 2022). In the service failure context, customers attribute more responsibility to anthropomorphized AI agents and perceive them as less stable, thus triggering more tolerance (Arikan et al., 2023).

However, alternative perspectives suggest that AI agent anthropomorphism may elicit negative responses. Customers have both explicit and implicit negative attitudes to highly anthropomorphized AI agents because of the threat of negative implications for humankind (Akdim et al., 2023). For example, Mende et al. (2019) find that customers interacting with anthropomorphized AI agents feel a sense of eeriness and perceive a threat to human identity, leading to more negative reactions that appear more anthropomorphized. Cheng (2023) also points out that customers are more likely to attribute blame for service failure to highly anthropomorphized AI agents, further decreasing failure tolerance.

Based on these findings, we propose that the relationship between AI agent anthropomorphism and failure tolerance is curvilinear, following an inverted U-shape. Specifically, at low to moderate levels of anthropomorphism, customers tend to have positive emotional responses, as indicated by the uncanny valley theory. According to the cognitive appraisal theory, these positive emotions can help alleviate the stress of service failure, thus increasing failure tolerance. However, as anthropomorphism increases from modest to high levels, AI agents may elicit negative emotional responses, also explained by the uncanny valley theory. Such negative emotions may exacerbate the stress from service failure, thus reducing customer's failure tolerance. Building on these arguments, we hypothesize that AI agent anthropomorphism promotes customer failure tolerance up to a certain threshold, beyond which it becomes counterproductive. Accordingly, we suggest the following hypothesis.

H1. The relationship between AI agent anthropomorphism and failure tolerance follows an inverted U-shape.

2.4. The inverted U-shaped of AI agent anthropomorphism and likability

Likability is a central element in the emotional connection formed in customer-AI interactions, reflecting customers' emotional disposition toward AI agents (Mathur and Reichling, 2016; McGuire, 1985). It serves as a key indicator of positive customer emotions and generally refers to the quality of being pleasant and appealing (Sandoval et al., 2021). While likability is influenced by anthropomorphism, this effect remains debated.

This research posits that the effect of AI agent anthropomorphism on customer perceived likability is curvilinear, stemming from the confounding effects associated with anthropomorphism. Customer cognition of anthropomorphism involves a dynamic process that may shift between self-identity affirmation and threat, depending on the degree of anthropomorphism (Blut et al., 2021; Mende et al., 2019; Yang et al., 2024). At low to modest levels of anthropomorphism, AI agents can enhance self-identity affirmation, as proposed by the uncanny valley theory (Blut et al., 2021; Mori, 1970). Through the insight of cognitive appraisal theory, this affirmation of identity increases a sense of comfort, increasing likability. Specifically, at a moderate degree of anthropomorphism, AI agents are more likely to fulfill customers' identity needs, evoking a greater sense of ease compared to non-anthropomorphic agents (Becker et al., 2023). This comfort is initially appraised by customers as a positive contribution to their personal well-being, thereby enhancing their perceived likability of the AI agent.

However, as anthropomorphism intensifies from modest to high levels, customers may begin to perceive the AI agent as a potential identity threat (Stein and Ohler, 2017). This perception, in turn, generates negative emotional responses, as customers view highly anthropomorphized AI agents as a potential threat to their sense of self and well-being. Cognitive appraisal theory suggests that such identity threats reduce likability, as customers appraise these agents as less compatible with their own identity and emotional needs. Consequently, we hypothesize the following.

H2. The relationship between AI agent anthropomorphism and likability follows an inverted U-shape.

2.5. The impact of likability on failure tolerance

Likability represents a fundamental aspect of human psychology, engendering positive emotional responses that influence various facets of customer intention and behavior (Nguyen et al., 2013), such as e-WOM intention (Jin and Hwang, 2024), revisit intention (Akarsu et al., 2020), usage intention (Cameron et al., 2021), recommend intention (Akarsu et al., 2020), attachment (Teng and Tsai, 2020), participation behaviors (Teng and Tsai, 2020), and other positive consequences. In the context of service failure, likability can serve as a cognitive buffer, mitigating negative emotions through a cognitive appraisal perspective (Lazarus and Folkman, 1984). Rather than simply experiencing dissatisfaction or disappointment, likable perceptions encourage customers to employ more constructive coping strategies in response to service failures. One such strategy is failure tolerance, a proactive approach that diminishes the negative impact of service failure on the firm (Lv et al., 2021). Customers with higher failure tolerance are more forgiving and tend to keep using the firm's services, thereby preserving customer satisfaction, loyalty, brand image, and other positive consequences (Zeithaml et al., 1993). Therefore, we argue that the likability can enhance their service failure tolerance.

H3. Likability positively influences failure tolerance.

2.6. The mediating role of likability

According to the cognitive appraisal theory, anthropomorphism influences failure tolerance by considering customers' perceived positive or negative emotional responses. When AI agents exhibit low levels of anthropomorphism, customers perceive minor likability from AI agents, which fails to sufficiently alleviate the negative emotions triggered by service failures. Consequently, failure tolerance remains low in such conditions. As the level of anthropomorphism increases, customers' perceived likability also increases. In the modest-anthropomorphic condition, the positive emotion derived from likability outweighs the negative emotion caused by service failure, leading to higher levels of failure tolerance. However, when anthropomorphism exceeds a certain threshold, customers experience the uncanny valley effect (Mori, 1970), wherein AI agents that appear too human-like provoke discomfort and reduce perceived likability. Despite the high anthropomorphism, the diminished likability at this stage leads to a reduction in failure tolerance, as customers' negative emotions are no longer effectively mitigated. Therefore, by combining H1, H2, and H3, we propose the following hypothesis for the role of likability.

H4. Likability mediates the inverted U-shaped effect of AI agent anthropomorphism on failure tolerance.

2.7. The moderating role of failure severity

To ensure the integrity of research results, it is essential to consider failure severity when examining service failure (Wang et al., 2011; Webster and Sundaram, 1998). Failure severity refers to the perceived intensity of a service problem, where a more severe failure is associated with greater perceived loss by the customer (Weun et al., 2004). Previous research highlights that failure severity affects customer responses to service failures. For instance, Park et al. (2023) find that failure severity negatively moderates the communication style effect on intimacy. Yang et al. (2022) demonstrate that failure severity can negatively moderate the influence of an agent's humor on customer evaluations of service failure. Specifically, this effect is more positive under low-severity conditions but becomes more negative under high-severity conditions. Similarly, Lv et al. (2021) suggest that in cases of low-severity failure, customers are more likely to tolerate failures by perceiving AI assistance as "cute"; however, in cases of high-severity failure, the positive influence of AI cuteness on failure tolerance diminishes.

Cognitive appraisal theory suggests that failure severity, as a cognitive construct, influences how customers assess the threat of loss posed by failure, thereby shaping their emotional responses to failure (Wang et al., 2011). When an AI agent possesses a modest level of anthropomorphism, customers typically experience positive emotional responses due to enhanced likability. However, high-severity failure elevates perceived loss threats, triggering negative emotions that can overshadow these positive effects. Specifically, in low-severity failures, customers perceive the incident as a minor inconvenience (Smith et al., 1999), which may lead to a more favorable view of the anthropomorphic qualities of the AI agent and allow positive emotions to influence their evaluation. Conversely, high-severity failures are more likely to be perceived as frustrating or threatening, leading to a heightened sense of loss and eliciting emotions such as anger or anxiety (Smith et al., 1999). Consequently, in high-severity scenarios, the initially positive emotional responses evoked by moderate anthropomorphism are often weakened by the intensity of negative emotions, thereby reducing the likability. Thus, we argue that service failure severity plays a moderating role in the curvilinear relationship between anthropomorphism and likability.

H5. Failure severity moderates the inverted U-shaped relationship between AI agent anthropomorphism and likability, attenuating (vs. amplifying) the effect under high (vs. low) failure severity.

Further, failure severity is closely associated with failure tolerance, affecting how customers behaviorally respond to service failures depending on the perceived intensity of the service failure. Under conditions of low failure severity, cognitive appraisal theory suggests that customers perceive a low-severity service failure as low stress, allowing positive emotions (i.e., likability) elicited by AI agent anthropomorphism to be more readily perceived. In this context, anthropomorphism's influence on failure tolerance is through the mediating effect of likability. As anthropomorphism increases from a low to a modest level, the uncanny valley theory posits that likability also increases, which in turn enhances failure tolerance. However, when anthropomorphism reaches a high level, customers may decline the likability and further reduce failure tolerance.

Under conditions of high failure severity, customers are likely to experience greater stress, perceiving the failure as a significant loss. According to the cognitive appraisal theory, such a perceived threat intensifies negative emotions, drawing attention away from the positive effects of likability associated with AI agent anthropomorphism. Thus, in high-severity situations, the influence of anthropomorphism on failure tolerance operates directly, without mediating the role of likability. Thus, we argue that failure severity is the moderator in the indirect relationship between anthropomorphism and failure tolerance via likability.

H6. Failure severity moderates the indirect relationship between AI agent anthropomorphism and failure tolerance through likability, attenuating (vs. amplifying) the mediation under high (vs. low) failure severity.

Fig. 1 illustrates the research's conceptual framework.

To test the proposed hypotheses, three laboratory experiments were designed. In each experiment, participants followed the experimental requirements to complete a given task. The first study used video scenarios to manipulate three levels of AI agent anthropomorphism and to validate stimulus material effectiveness. The second study used text scenarios to manipulate service failure severity and to validate the stimulus validity of the experimental material. These two preliminary studies laid the groundwork for subsequent main experiments involving stimulus material. In the third study, we conducted a video vignette-based experiment to test all hypotheses.

3. Study 1

This study aims to investigate the effectiveness and validity of anthropomorphism manipulations in AI agents within customer service scenarios. To achieve this goal, we ensure that all other variables remain constant while varying only the degree of anthropomorphism in the AI agents. It should be clarified that in this study, AI agents refer to physical entities that utilize automation technologies to communicate and interact with customers.

3.1. Stimulus materials

We focused particularly on the anthropomorphic characteristics of the AI agents and utilized an online experimental platform where participants watched a video. The videos illustrated the interaction between customers and AI agents. After watching the video, participants were instructed to assess the degree of anthropomorphism in the AI agents based on the content they observed. To maintain the reliability and consistency of the experimental outcomes, we adopted standardized video scripts, ensuring that identical information was presented across all levels of AI agent anthropomorphism. The scripts depicted real hotel front desk scenarios, with anthropomorphism manipulated by adjusting the human-like attributes of the AI agents. All three video scripts were synthesized using professional video editing software based on real-life scenarios, ensuring continuity and authenticity in the scenes and processes. Additionally, to enhance the immersion of the experiment, we required participants to wear headphones before the experiment started and ensured that the videos were played in full-screen mode to minimize external distractions. This setup helped participants focus more on the video content and the anthropomorphic performance of the AI agents.

3.2. Samples

Participants were recruited by purposive sampling from the Credamo (https://www.credamo.com), a professional online experiment platform with access to a diverse panel of over three million participants. Widely utilized in research across disciplines such as psychology, consumer behavior, and organizational studies, Credamo ensures a broad and representative pool of respondents for surveys and experiments. This platform allows researchers to set specific screening criteria to target relevant populations, enhancing the validity and applicability of the study's findings. Ninety-six participants (57.3 % female, $M_{age} = 31.7$) with travel experience in the past year and using hotel service agents were recruited. According to the calculation of G*Power, the sample size surpassed the requirement for adequate statistical power to identify medium effect sizes (Cohen, 2013). The detailed demographic information is presented in Table 1.

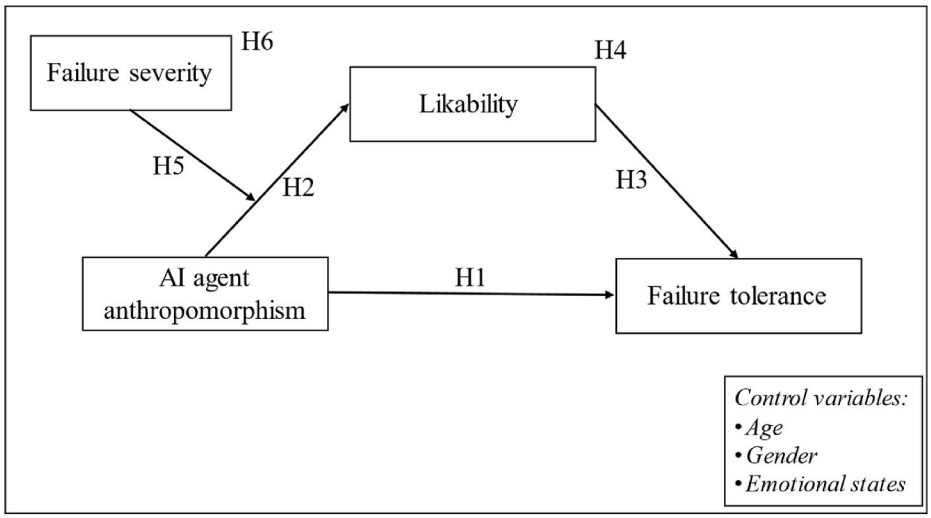


Fig. 1. Conceptual framework of the research.

Table 1
Participant demographic information.

Variable	Item	Study 1		Study 2		Study 3	
		Frequency	Percent (%)	Frequency	Percent (%)	Frequency	Percent (%)
Gender	Male	41	42.7	24	30.0	160	45.7
	Female	55	57.3	56	70.0	190	54.3
Age	≤20	20	20.8	15	18.8	33	9.4
	21–30	49	51.0	43	53.8	156	44.6
	31–40	20	20.8	17	21.3	128	36.6
	41–50	5	5.2	2	2.5	21	6.0
	≥51	2	2.1	3	3.8	12	3.4

3.3. Procedure

Participants were randomly assigned to one of three conditions, each representing a distinct level of AI agent anthropomorphism. Participants were required to use a computer to access the experimental system. Upon entering the system, they received experimental instructions informing them of the upcoming video content. Additionally, an on-screen prompt instructed participants to watch the video attentively, as subsequent questionnaire questions would be based on the video content. The video was set to automatically play in full-screen mode, and participants were not allowed to exit the experiment midway. In the video, the AI agent first interacted with the customer, greeted and introduced itself as an AI agent. The AI agent then inquired about the customer's needs and, upon receiving a specific request, indicated that it would address the request immediately. To eliminate the potential impact of service success or failure on the anthropomorphism evaluation, the video ended after the AI agent responded to the customer's request. This design ensured that the evaluation focused only on the anthropomorphic characteristics of the AI agent, rather than the service outcome. After the video, the participants' experimental interface automatically transitioned to the questionnaire section in which measured the AI agent anthropomorphism and collected demographic information. Upon completing the experiment, each participant received a cash reward. The experimental procedures are shown in Fig. 2.

3.4. Measures

To ensure the randomness of the group assignments, we utilized the professional automatic randomization function provided by the Credamo platform. The Chi-square tests for gender ($\chi 2$ (2, $N = 96$) = 0.596, $p = 0.742$) and age distribution ($\chi 2$ (8, $N = 96$) = 6.331, $p = 0.610$) indicate no significant difference across the control and experimental groups, supporting the effectiveness of our randomization process.

To verify the effectiveness of our anthropomorphism manipulation, we measured participants' perceived anthropomorphism using a continuous slider scale from 1 to 100 ("How human-like do you perceive this AI agent to be? Please indicate by dragging the slider, where '1' represents completely non-human and '100' represents extremely human-like").

3.5. Results

To validate the anthropomorphic differences between the three AI agents, we performed a one-way ANOVA accompanied by a post-hoc test. The one-way ANOVA analysis showed that there was a significant difference between the anthropomorphism of the three agents ($M_{high} = 88.78$, $SD_{high} = 4.89$, $M_{modest} = 75.25$, $SD_{modest} = 14.22$, $M_{low} = 24.81$, $SD_{low} = 27.55$, F (2,93) = 110.797, $p < 0.001$). Furthermore, the post-hoc comparisons with Fisher's Least Significant Difference (LSD) test indicate that the high-anthropomorphism AI agent had the highest level of anthropomorphism and was significantly higher than the modest ($p < 0.01$) and the low ones ($p < 0.001$), respectively. The level of anthropomorphism of the modest-anthropomorphism agent was significantly higher than the low one ($p < 0.001$). Therefore, the anthropomorphism stimuli designed in this study were effective.

4. Study 2

The purpose of this study is to verify the effectiveness and validity of manipulating two levels of failure severity in service scenarios.

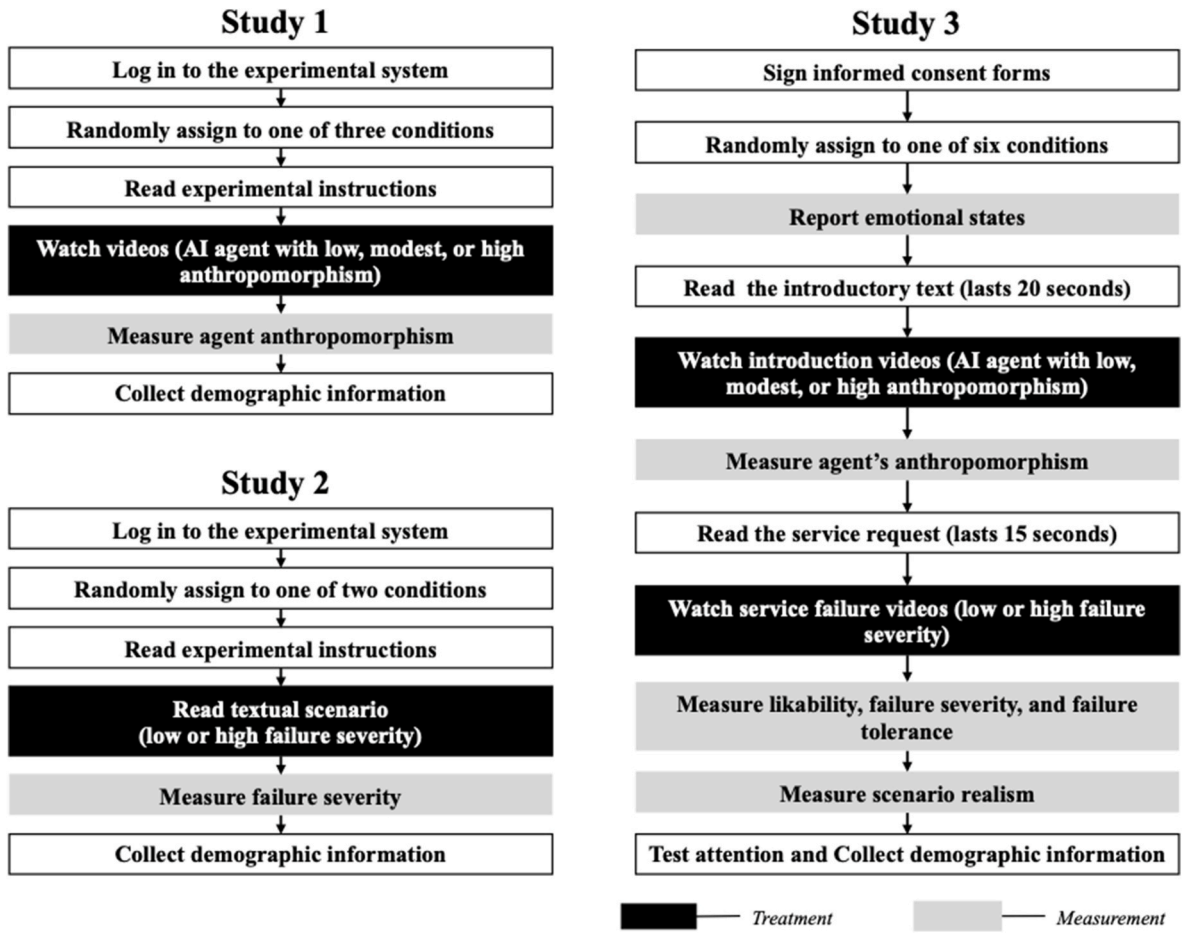


Fig. 2. Experimental design for Studies 1, 2, and 3.

4.1. Stimulus materials

We adopted a service scenario from Smith et al. (1999) involving hotel check-in, presented in a text-based format. We pre-designed the textual scripts for the scenarios, which were repeatedly reviewed and revised by two experts to ensure linguistic fluency and scenario authenticity. In the two scenarios, except for the difference in severity, all other factors, such as the service provider, service process, event location, and event timing, remained identical.

4.2. Samples

Participants were recruited by purposive sampling from the Credamo platform. Eighty participants (70.0 % female, $M_{age} = 27.4$) with travel experience in the past year and using hotel service agents were recruited. The sample size was above the threshold needed to achieve sufficient statistical power for detecting medium effect sizes (Cohen, 2013). The detailed demographic information is shown in Table 1.

4.3. Procedure

Similar to Study 1, participants were randomly assigned to one of two conditions, each representing a low or high level of failure severity condition. Participants were required to use a computer to access the experimental system. Upon logging into the experimental platform, participants were randomly assigned to either the low-severity scenario or the high-severity scenario. Subsequently, the participant's computer interface displays a text scenario (see Table 2). After reading the textual scenarios, participants clicked a button to proceed to the questionnaire section which measured the service failure severity and collected demographic information. Upon completing the experiment, each participant received a cash reward. Fig. 2 illustrates the experimental procedure.

Table 2
Text-based scenarios for low or high failure severity scenarios.

Low-severity Condition	High-severity Condition
Imagine yourself in the following situation:	Imagine yourself in the following situation:
At 7 PM, you arrive at the hotel you booked for your important trip to check in at the front desk. The AI agent at the front desk will serve you …	At 7 PM, you arrive at the hotel you booked for your important trip to check in at the front desk. The AI agent at the front desk will serve you …
The AI agent says, "Hello, how can I help you?"	The AI agent says, "Hello, how can I help you?"
You tell the AI agent, "I booked a room online and would like to check in now, thank you."	You tell the AI agent, "I booked a room online and would like to check in now, thank you."
The AI agent says, "I'm glad to help. Let me check your reservation." (One minute later)	The AI agent says, "I'm glad to help. Let me check your reservation." (One minute later)
The AI agent says, "I'm sorry, we are overbooked today and the room you reserved is no longer available. However, we can offer you a different type of room."	The AI agent says, "I'm sorry, we are overbooked today and the room you reserved is no longer available. We currently have no available rooms to offer you."
At this moment, you realize that the AI agent has not met your needs, representing a service failure.	At this moment, you realize that the AI agent has not met your needs, representing a service failure.

4.4. Measures

We adapted Hess's (2008) 5-point Likert scale to measure participants' evaluations of service failure severity in the experimental scenarios (1 = "Strongly disagree" and 5 = "Strongly agree"). This scale included three items: "To what extent do you perceive the seriousness of this mistake made by the agent?", "To what extent was the mistake made by the agent inconvenient to you?", and "To what extent do you consider this situation fair?" ($\alpha = 0.837$). After measuring the failure severity, participants were asked to report their age and gender information.

4.5. Results

We used an independent samples *t*-test to verify that there was a difference in perceived severity between the two failure scenarios. The results showed a higher severity perception in the high-severity condition ($M = 3.82$, $SD = 0.74$) compared to the low-severity condition ($M = 2.85$, $SD = 0.79$, $t(78) = 5.67$, $p < 0.001$). Therefore, the failure severity stimuli were effective.

5. Study 3

This study aims to provide initial evidence for the curvilinear relationship between anthropomorphism and failure tolerance through likability and the relationship boundary of failure severity through a 3 (anthropomorphism: low vs. modest vs. high) × 2(failure severity: low vs. high) between-subjects factorial design based on video vignettes widely employed in customer behavior research.

5.1. Stimulus materials

We totally produced 9 pieces of video vignettes to feature a check-in service encounter, including 3 pieces of video vignettes of greeting and service introductions from three anthropomorphic AI agents and 6 pieces of video vignettes of service failure encounters from different anthropomorphic AI agents in different failure severity. The experimental materials used in the video have been validated as effective by two previous studies, including three AI agents with different levels of anthropomorphism and two service failure scenarios with different levels of severity. Then, we invited three experts from different research fields to read and revise the scripts word by word to ensure the validity and reliability of video vignettes (Chen et al., 2022). According to the scripts, we use professional video editing software to produce video vignettes.

5.2. Samples

Participants were recruited by purposive sampling through the Credamo platform. A total of 350 participants (54.3 % female; $M_{age} = 32.9$) with travel experience in the past year and using hotel service agents were recruited. The sample size far exceeded sufficient statistical power to detect medium effect sizes (Cohen, 2013). The detailed demographic information is illustrated in Table 1. All participants signed informed consent forms, understood the potential benefits and risks of the study, and willingly participated in the experiment.

Before the beginning of the experiment, participants were asked to be in a quiet environment and to use computer equipment with an Internet connection. In addition, to ensure an optimal audio-visual experience, participants were asked to wear headphones and adjust the volume to the appropriate level. Participants were also informed that to ensure concentration, the experimental program would be automatically full-screened on the computer during the experiment, and it would not be possible to continue if they exited during the experiment.

5.3. Procedure

At the beginning of the experiment, participants were randomly assigned to one of six conditions (3 anthropomorphism × 2 failure severity). Then, participants reported emotional states to control for effects on likability and failure tolerance. Subsequently, the experimental interface showed that "Now, you are on an important trip. At 7 p.m., you arrive at the hotel you booked for your important trip to check in at the front desk. The AI agent at the front desk will serve you …" 20 s later, the experimental program automatically played one of the three service introduction videos of the AI agent with low, modest, or high anthropomorphism, in which the AI agent said "Hello, I am the AI agent here, how can I help you?" After that video, participants measured the agent's anthropomorphism. Then the screen displayed the service request that asked the AI agent to please check in according to the reservation.

After 15 s, the next video was automatically played in which the AI agent received requirements and provided a failed service (with low or high severity) about an overbooking situation. In the high-severity condition, the AI agent said "I'm glad to help! Let me check your reservation. (1 min later) I'm sorry, we are overbooked today and the room you reserved is no longer available. We currently have no available rooms to offer you." In the low-severity condition, the AI agent said "I'm glad to help! Let me check your reservation. (1 min later) I'm sorry, we are overbooked today and the room you reserved is no longer available. However, we can offer you a different type of room" After the end of the video, the experimental interface displayed "Please answer the following questions according to the video scenario", and the participants clicked on the "Continue" button to skip to the questionnaire page. In the next questionnaire, likability, failure severity, failure tolerance, and scenario realism were measured in order using scales. Additionally, participants were asked to report their demographic information and recall specific details of the service scenario as an attention check and video-vignette reliability and validity confirmation (Chen et al., 2022). Finally, a cash bonus was awarded. The procedures of this experiment are shown in Fig. 2.

5.4. Measures

AI agent anthropomorphism was measured using a continuous slider from 1 to 100, which was the same as the measuring instrument employed in Study 1. Likability was measured using the 5-point semantic differential scale with 5 items of Bartneck et al. (2009) (dislike-like; unfriendly-friendly; unkind-kind; unpleasant-pleasant; awful-nice). Depending on the scenario displayed in the video, participants then reported the perceived failure severity using the same scale as in study 2 and reported failure tolerance on a 5-point Likert scale with 4 items by Augusto de Matos et al. (2009) ("This failure is deemed forgivable."; "You would refrain from sharing this negative experience with others."; "You would be willing to request the AI agent to redo the command."; "You would consider recommending the AI agent to others.", 1 = "strongly disagree" and 5 = "strongly agree"). The scenario realism was measured using Sparks and Browning's (2011) 7-point scale with 5 items ("I perceive this service failure scenario as authentic."; "I can envisage encountering this situation during a hotel check-in."; "I find this situation analogous to real-life occurrences."; "I believe such incidents do occur in hotels."; "I deem it realistic and plausible to experience this situation in a hotel.", 1 = "strongly disagree" and 7 = "strongly agree", $\alpha = 0.871$). Emotional state was measured by Townsend and Sood's (2012) five-point semantic differential scale (sad--happy; bad mood-good mood; irritable-pleased; depressed-cheerful). According to Podsakoff et al. (2003), different scale endpoints and anchors were employed to reduce common method biases.

5.5. Reliability and validity assessment

The reliability and validity of the scales were evaluated. The Cronbach's alpha values for all scales were above 0.8 (see Table 3), indicating internal consistency. To assess convergent validity, we used composite reliability (CR) and average variance extracted (AVE). As shown in Table 3, all CR values exceed 0.8, suggesting that the measures are reliable. The AVE values range from 0.582 to 0.765, all exceeding the threshold of 0.5, providing evidence of convergent validity. Additionally, the factor loadings for all constructs were above the strong correlation threshold of 0.5, further confirming the convergent validity. Table 4 presents the square root of AVE and the correlation coefficients. Since the square root of each AVE is greater than the correlation coefficients with other constructs, discriminant validity is well established.

The confirmatory factor analysis demonstrated that the model was well-fitted ($\chi 2/df = 1.684$, RMSEA = 0.044, GFI = 0.943, CFI = 0.983, NFI = 0.959, AGFI = 0.921). These results indicate that the measurement model adequately represents the data.

Endogeneity is a concern in research involving self-reported measures, as it may arise from omitted variables, common method bias (CMB), measurement error, and simultaneity (Antonakis et al., 2010). To reduce the risk of omitted variable bias, we included age, gender, and emotional state as control variables in the analysis. These variables were selected based on their relevance to the relationship between AI agent anthropomorphism and failure tolerance, as they may influence participants' perceptions and responses during the experiment (Lv et al., 2021). The reliability and validity of the questionnaire-based measures were rigorously evaluated to address measurement errors. Simultaneity is less likely to be a concern in an experimental context, as the causal direction between variables is controlled through random assignment of conditions. This design feature ensures that any observed effects can be attributed to the experimental manipulations rather than reverse causality.

Although experiments inherently reduce some risks of common method biases (e.g., by manipulating independent variables rather than relying solely on self-report), the use of questionnaire-based dependent measures could still introduce common method biases. Accordingly, we conducted Harman's single-factor test using exploratory factor analysis (EFA). The results showed that the largest single factor explained 31.22 % of the total variance, which is well below the 50 % threshold, indicating that common method biases are unlikely to significantly impact the results.

The variance inflation factor (VIF) test was conducted to evaluate multicollinearity among the predictors. The results are presented in Table 5. In this study, all VIF values are well below the commonly accepted threshold of 4, with the maximum VIF being 1.300. Additionally, the tolerance values for all predictors exceed the minimum threshold of 0.1, further confirming that multicollinearity is not a concern.

Table 3
Reliability and convergent validity.

Variables	Items	Factor Loadings	Cronbach's α	CR	AVE
Likability	LB1	0.897	0.941	0.942	0.765
	LB2	0.860			
	LB3	0.791			
	LB4	0.926			
	LB5	0.893			
Failure severity	FS1	0.814	0.885	0.888	0.725
	FS2	0.891			
	FS3	0.848			
Failure tolerance	FT1	0.839	0.837	0.845	0.582
	FT2	0.823			
	FT3	0.558			
	FT4	0.797			
Emotional state	ES1	0.861	0.921	0.923	0.749
	ES2	0.829			
	ES3	0.881			
	ES4	0.890			

Table 4
Discriminant validity and correlations in Study 3.

Variables	1	2	3	4
1. Likability	0.874			
2. Failure severity	−0.203	0.852		
3. Failure tolerance	0.213	−0.281	0.763	
4. Emotional state	0.199	−0.148	−0.058	0.866

Notes: The values on the diagonal are the square root of the AVE.

Table 5
Multicollinearity diagnostics for predictors of failure tolerance and likability.

Dependent variables	Predictors	VIF	Tolerance	Results
Failure tolerance	Gender	1.075	0.930	No multicollinearity
	Age	1.096	0.912	
	Emotional state	1.103	0.907	
	Anthropomorphism	1.189	0.841	
	Likability	1.300	0.769	
	Failure severity	1.066	0.938	
Likability	Gender	1.050	0.952	No multicollinearity
	Age	1.090	0.917	
	Emotional state	1.065	0.939	
	Anthropomorphism	1.019	0.981	
	Failure severity	1.047	0.955	

6. Results

6.1. Manipulation check

The results of one-way ANOVA with post-hoc comparisons with Fisher's Least Significant Difference (LSD) test showed significant differences in anthropomorphism ($M_{low} = 25.67$, $SD_{low} = 25.74$, $M_{modest} = 54.74$, $SD_{modest} = 28.58$, $M_{high} = 62.42$, $SD_{high} = 21.16$, $F\ (2,347) = 72.01$, $p < 0.001$), indicating the manipulation of anthropomorphism was effective. The failure severity reported by participants was significantly different ($M_{low} = 2.74$, $SD_{low} = 0.97$, $M_{high} = 3.68$, $SD_{high} = 0.99$, $t\ (348) = -9.04$, $p < 0.001$), indicating the manipulation of failure severity was also effective. The scenarios in the experiment were more realistic than the mean value ($M = 5.72$, $SD = 0.84$, $t\ (349) = 49.426$, $p < 0.001$), indicating the realism of the scenarios and the validity of the videos. Thus, both stimuli and realism manipulation were considered effective.

6.2. Hypothesis test

We used one-way ANCOVA and hierarchical linear regression to comprehensively test the hypotheses concerning curvilinear relationships, supplemented by bootstrapping analyses to examine indirect relationship hypotheses. Table 6 presents the descriptive statistics for the main variables in the model. According to Tabachnick and Fidell (2013), data is considered acceptable for normality if the absolute value of skewness is less than 2 and the absolute value of kurtosis is less than 7. These results satisfy the distributional assumptions required for hypothesis testing.

6.3. Curvilinear relationships

The results of one-way ANCOVA revealed that there was a significant difference in likability ($F\ (2, 344) = 90.972$, $p < 0.001$) and failure tolerance ($F\ (2, 344) = 9.409$, $p < 0.001$) under different levels of anthropomorphism, controlling for age, gender, and emotional state.

Table 6
Descriptive statistics for the main variables.

	N	Min.	Max.	Mean	SD	Skewness	Kurtosis
AI agent anthropomorphism	350	0.000	2.000	0.960	0.833	0.081	−1.556
Likability	350	1.000	5.000	3.663	1.056	−0.826	−0.577
Failure tolerance	350	1.000	5.000	3.036	0.961	−0.053	−0.997
Failure severity	350	0.000	1.000	0.480	0.500	0.080	−2.005
Emotional state	350	1.500	7.000	5.549	1.186	−1.062	0.580
Age	350	1.000	5.000	2.490	0.875	0.636	0.761
Gender	350	0.000	1.000	0.460	0.499	0.173	−1.981

Specifically, participants within the modest-anthropomorphism group exhibited significantly greater failure tolerance (M_{modest} = 3.35, SD_{modest} = 0.97) compared to both the low-anthropomorphism (M_{low} = 2.81, SD_{low} = 0.93, t (234) = 4.319, $p < 0.001$) and high-anthropomorphism (M_{high} = 3.00, SD_{high} = 0.92, t (219) = 2.707, $p < 0.01$), providing preliminary support for H1. Similarly, participants in the modest-anthropomorphism group (M_{modest} = 4.42, SD_{modest} = 0.41) demonstrated significantly higher likability relative to both the low-anthropomorphism group (M_{low} = 2.90, SD_{low} = 1.09, t (234) = 13.624, $p < 0.001$) and high-anthropomorphism group (M_{high} = 3.82, SD_{high} = 0.85, t (164.380) = 6.629, $p < 0.001$), initially supporting H2.

Furthermore, hierarchical linear regression, utilizing direct scores of anthropomorphisms (0 = low, 1 = modest, 2 = high), was employed to further validate the hypothesized inverted U-shaped curvilinear relationship. Considering that different endpoint scales were used to measure the variables in the study, the data were standardized before conducting the regression analysis. The results of the standardized model are presented in Tables 7 and 8, while additional information about the unstandardized model, including unstandardized coefficients, standard errors, t-values, and p-values, is provided in Appendix A. Model 3 highlighted a significantly positive main effect of anthropomorphism on failure tolerance (β = 0.847, $p < 0.001$), and a significantly negative coefficient for the anthropomorphism quadratic term (β = −0.790, $p < 0.001$), supporting H1. The impact of anthropomorphism on failure tolerance follows a pattern of gradual increase followed by a gradual decrease (see Fig. 3, Panel A). Similarly, Model 9 revealed a significantly positive main effect of anthropomorphism on likability (β = 1.914, $p < 0.001$), while the regression coefficient for the quadratic term of anthropomorphism was significantly negative (β = −1.598, $p < 0.001$), supporting H2. This suggests that the effect of anthropomorphism on likability increases up to the inflection point and then decreases thereafter (see Fig. 3, Panel B). Additionally, Model 6 demonstrated that likability positively influenced failure tolerance (β = 0.225, $p < 0.001$), supporting H3.

6.3.1. Indirect effect analysis

Bootstrapping analyses (utilizing the MEDCURVE method with 10,000 bootstrapping samples; Hayes and Preacher, 2010) examined the instantaneous indirect effect of anthropomorphism on failure tolerance, mediated by likability. The effect was significantly positive at the relatively low level of anthropomorphism (θ = 0.249, SE = 0.121, 95 % CI: [0.017, 0.490]) and positive at the relatively modest level of anthropomorphism (θ = 0.062, SE = 0.031, 95 % CI: [0.005, 0.126]). However, at the relatively high level of anthropomorphism, this effect reversed, becoming significantly negative (θ = −0.125, SE = 0.062, 95 % CI: [−0.251, −0.009]). Therefore, likability mediated the inverted U-shaped impact of AI agent anthropomorphism on failure tolerance, supporting H4.

6.3.2. Moderation analysis

Hierarchical linear regression analysis, specifically Model 12, highlighted a significant positive impact of the interaction term between anthropomorphism quadratic term and failure severity on likability (β = 0.565, $p < 0.05$). As shown in Fig. 4, when confronted with high-severity failure, significant differences in likability emerged between low- and modest-anthropomorphism conditions (M_{low} = 3.07, SD_{low} = 1.092, M_{modest} = 4.27, SD_{modest} = 0.47, t (87.78) = −7.898, $p < 0.001$). Similarly, a significant difference was observed between modest- and high-anthropomorphism conditions (M_{modest} = 4.27, SD_{modest} = 0.47, M_{high} = 3.76, SD_{high} = 0.85, t (81.40) = 3.860, $p < 0.001$). When failure severity was low, there were significant differences in likability between the low- and modest-anthropomorphism (M_{low} = 2.74, SD_{low} = 1.07, M_{modest} = 4.55, SD_{modest} = 0.27, t (74.916) = −13.251, $p < 0.001$), as well as between modest- and high-anthropomorphism conditions (M_{modest} = 4.55, SD_{modest} = 0.27, M_{high} = 3.87, SD_{high} = 0.85, t (73.450) = 5.945, $p < 0.001$). Therefore, the results suggested that failure severity moderated the curvilinear relationship between anthropomorphism and likability and such a relationship was attenuated (vs. amplified) in high (vs. low) failure severity, supporting H5.

Table 7
Hierarchical linear regression on failure tolerance (standardized).

Variables	Failure tolerance					
	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6
Control variables						
Age	0.051	0.043	0.033	0.026	0.010	0.027
Gender	0.06	0.061	0.039	0.023	0.022	0.026
Emotional state	−0.07	−0.067	−0.086	−0.108*	−0.100	−0.109*
Independent variables						
Anthropomorphism		0.085	0.847***	0.571*	0.416	
Anthropomorphism squared			−0.790***	−0.560*	−0.527	
Failure severity					−0.345^{+}	
Interaction terms						
Anthropomorphism × Failure severity					0.223	
Anthropomorphism squared × Failure severity					−0.009	
Mediator						
Likability				0.144*	0.162*	0.225***
Total R^2	0.011	0.018	0.062	0.074	0.131	0.057
F-value	1.237	1.567	4.541***	4.600***	5.688***	5.258***

Notes: $^{+}p < 0.1$, $*p < 0.05$, $**p < 0.01$, $***p < 0.001$.

Table 8
Hierarchical linear regression on likability (standardized).

Variables	Likability					
	Model 7	Model 8	Model 9	Model 10	Model 11	Model 12
Control variables						
Age	0.105	0.07	0.051	0.051	0.056	0.055
Gender	0.149**	0.154**	0.109*	0.110*	0.105*	0.110*
Emotional state	0.175***	0.188***	0.149***	0.149***	0.142***	0.143***
Independent variables						
Anthropomorphism		0.374***	1.914***	1.914***	1.993***	2.309***
Anthropomorphism squared			−1.598***	−1.598***	−1.605***	−1.933***
Failure severity				0.008	0.095	0.152*
Interaction terms						
Anthropomorphism × Failure severity					−0.135^{+}	−0.722**
Anthropomorphism squared × Failure severity						0.565*
Total R^2	0.078	0.217	0.397	0.397	0.403	0.411
F-value	9.805***	23.894***	45.331***	37.677***	32.968***	29.784***

Notes: $^{+}p < 0.1$, $*p < 0.05$, $**p < 0.01$, $***p < 0.001$.

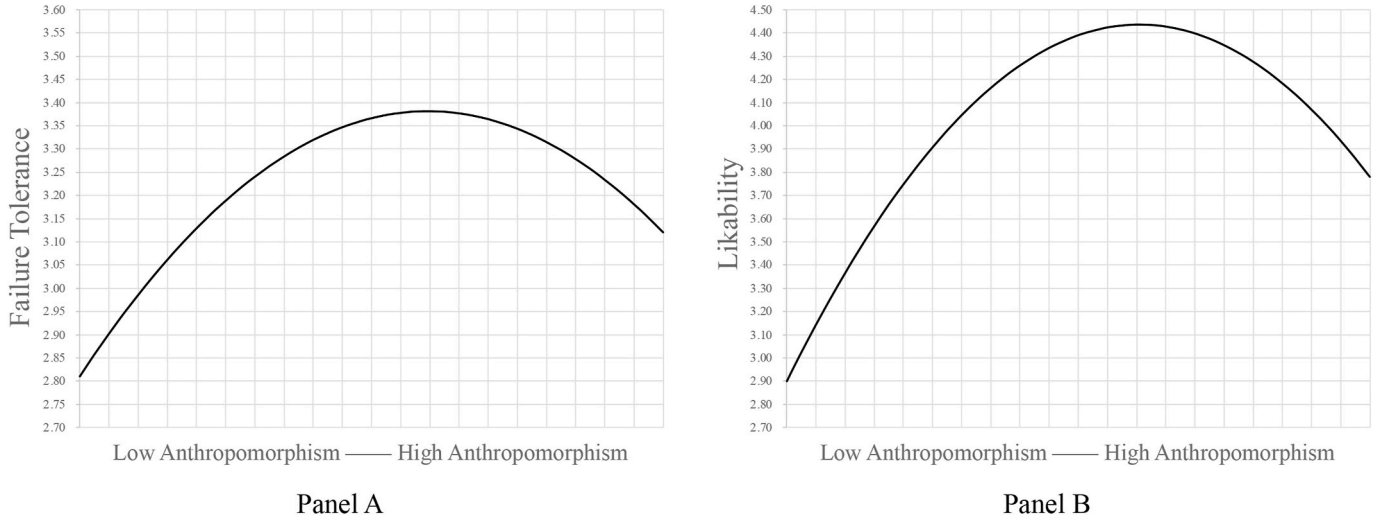


Fig. 3. Anthropomorphism's inverted U-shaped impact on failure tolerance (Panel A)/likability (Panel B).

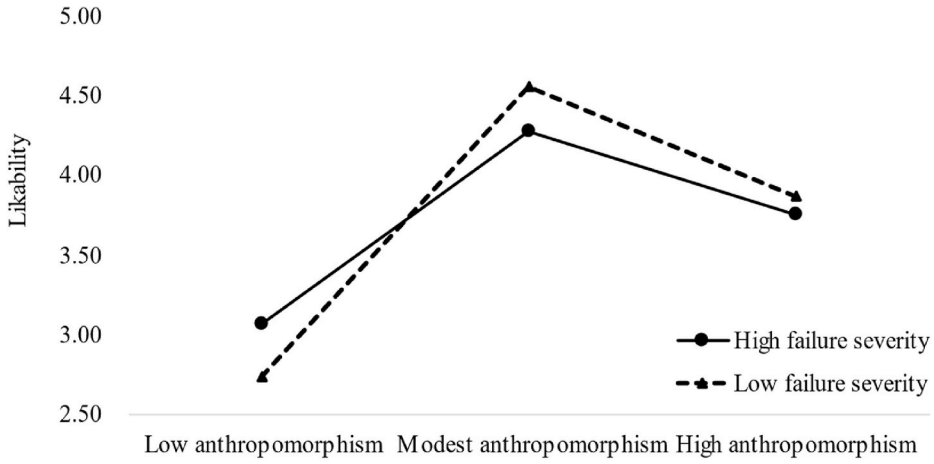


Fig. 4. The interactive effect of anthropomorphism and failure severity on likability.

6.3.3. Conditional indirect effect analysis

Bootstrapping analyses (MEDCURVE with 10,000 bootstrapping samples; Hayes and Preacher, 2010) revealed a conditional indirect effect, demonstrating the moderation of the indirect pathway from anthropomorphism to failure tolerance through likability by failure severity. Under the low-severity condition, the instantaneous indirect effect was increasingly positive and significant from relatively low anthropomorphism ($\theta = 0.541$, $SE = 0.209$, 95 % CI: [0.137, 0.963]) to the relatively modest level of anthropomorphism ($\theta = 0.127$, $SE = 0.055$, 95 % CI: [0.032, 0.251]), and became significantly negative at the relatively high level of anthropomorphism ($\theta = -0.288$, $SE = 0.106$, 95 % CI: [−0.498, −0.080]). However, under the high-severity

condition, there was insignificant instantaneous indirect effect either from relatively low anthropomorphism ($\theta = 0.089$, $SE = 0.127$, 95 % CI: [−0.147, 0.357]) to relatively modest level of anthropomorphism ($\theta = 0.025$, $SE = 0.035$, 95 % CI: [−0.043, 0.099]), or from modest level to relatively high level of anthropomorphism ($\theta = -0.039$, $SE = 0.059$, 95 % CI: [−0.174, 0.061]). Therefore, the indirect relationship between anthropomorphism and failure tolerance through likability is only significant at the low-severity condition, supporting H6.

7. Discussion and conclusions

7.1. Discussion

This research investigates how AI agent anthropomorphism influences failure tolerance in the context of service failures. The findings reveal an inverted U-shaped effect of AI agent anthropomorphism on failure tolerance, with likability as a mediator and failure severity as a moderator. These results provide insights into the mechanisms by which anthropomorphism influences consumer responses in failure contexts.

Challenging the traditional view that anthropomorphism always leads to positive customer outcomes, our findings reveal a curvilinear relationship. As AI agent anthropomorphism increases, both failure tolerance and likability initially rise but decline when anthropomorphism becomes excessive. While previous research predominantly highlights the benefits of anthropomorphism in customer experiences (Akdim et al., 2023; Arikan et al., 2023; Shehawy et al., 2025; Yang et al., 2024; Zheng et al., 2025), our results indicate that an overly anthropomorphized AI agent can instead trigger negative reactions. This phenomenon aligns with the uncanny valley theory (Mori, 1970; Söderlund, 2022), which posits that when human-like features exceed a certain threshold, they can elicit discomfort and reduce user satisfaction. Extending prior research on the drawbacks of extreme anthropomorphism (Blut et al., 2021; Mende et al., 2019; Yang et al., 2022), this study provides new insights into how AI agent anthropomorphism influences customer responses in service failure context.

Furthermore, our findings reveal that likability serves as a mediator in the curvilinear relationship between anthropomorphism and failure tolerance, which complements prior literature. While Blut et al. (2021) suggested that warmth, social presence, and animacy could mediate anthropomorphism's effect on customer behavior, they did not provide empirical validation for likability as a mediator. Our research fills this gap by demonstrating that likability indeed mediates this relationship, but only in a curvilinear pattern. Specifically, at low and moderate levels of anthropomorphism, likability positively mediates the relationship between anthropomorphism and failure tolerance, but at high anthropomorphism levels, a decline in likability reduces failure tolerance. This extends previous studies by offering empirical evidence on the emotional mechanisms through which anthropomorphism influences customer behavior.

Beyond its mediating role, our results also demonstrate that likability consistently exerts a positive direct effect on failure tolerance, regardless of the level of anthropomorphism. This finding reinforces prior research that identified positive affect as a key driver of customer behaviors (Cameron et al., 2021; Li et al., 2019; Jin and Hwang, 2024). However, this study extends this line of discussion by showing that even in high anthropomorphism conditions, where likability decreases, its direct positive influence on failure tolerance remains robust. Thus, likability not only mediates the effect of anthropomorphism on failure tolerance but also independently influences customer responses to service failures, highlighting its critical role in shaping customer behavior in human-AI interactions.

In contrast to previous research on anthropomorphism and customer emotions, which has primarily focused on factors such as failure type (Choi et al., 2021), empathy (Mehmood et al., 2024; Yang et al., 2024), and personalization (Cui and Zhong, 2023; Xie and Lei, 2022), this study identifies failure severity as a key boundary condition. Specifically, we demonstrate that failure severity weakens the curvilinear effect of AI agent anthropomorphism on likability. The anthropomorphism-likability curve shows a steeper incline in low-severity failure situations and flattens in high-severity contexts. This relationship has not been systematically examined in previous literature. While Akdim et al. (2023) argued that customers' emotional responses to service failures depend on failure severity, they did not explore how failure severity alters the anthropomorphism-likability relationship. Our findings provide empirical support for this proposition and expand the discussion by showing that in high-severity failures, customers are less sensitive to anthropomorphism variations, making anthropomorphism a less influential factor in shaping likability.

Finally, unlike previous studies that have focused on failure severity as a boundary condition for the main effect (Lv et al., 2021; Park et al., 2023), this study reveals that failure severity also moderates the mediating relationship. Failure severity moderates the mediation effect of likability on the relationship between anthropomorphism and failure tolerance. Under low-severity conditions, likability mediates this relationship in a curvilinear pattern, where anthropomorphism initially enhances and then diminishes failure tolerance. In high-severity scenarios, however, this mediated relationship disappears, with no significant indirect effect of anthropomorphism on failure tolerance. Thus, the indirect impact of anthropomorphism on failure tolerance via likability is evident only in low-severity failure situations. This finding aligns with previous research focusing on failure severity as a boundary condition, which similarly concludes that the effect is only significant under low-severity failure conditions (Lv et al., 2021; Park et al., 2023).

7.2. Theoretical implications

This research contributes to the AI agent anthropomorphism and service failure literature, identifies the mechanisms of the anthropomorphism-tolerance relationship, and extends the cognitive appraisal theory and uncanny valley theory.

First, in contrast to prior research, this paper uncovers the inverted U-shaped effect of AI agent anthropomorphism on failure tolerance. This resolves the apparent contradiction in the literature where service agent anthropomorphism has been posited to both enhance (e.g., Arikan et al., 2023; Zheng et al., 2025) and diminish (e.g., Cheng, 2023; Cui and Zhong, 2023; Zhang et al., 2023) failure tolerance. We demonstrate that modest levels of anthropomorphism, characterized by a balance between human likeness and machine distinctiveness, are optimal for eliciting customers' failure tolerance. This finding refines the emotional and behavioral implications of previous research on the curvilinear effects of anthropomorphism on customer responses (e.g., Lu et al., 2021; Xie and Lei, 2022; Yang et al., 2024), and extends our comprehension of the inverted U-shaped role of AI agent anthropomorphism in situations of service failure scenarios.

Further, this study identifies an internal emotional mechanism and a boundary condition within the intricate relationship between AI agent anthropomorphism and failure tolerance. Specifically, AI agent anthropomorphism indirectly influences failure tolerance through likability. However, this indirect effect is influenced by failure severity. Such effect is significant only in low-severity failure scenarios, whereas it diminishes in high-severity failures. This finding highlights the nuanced interaction between failure severity and the emotional responses triggered by anthropomorphism. Specifically, in cases of high service failure, anthropomorphic attributes may be perceived as inappropriately or incongruously applied, thus significantly reducing the tendency to elicit positive emotions. As a result, the activation of likable emotions is limited. This insight deepens our understanding of the role of anthropomorphism in service failure contexts, and extends the existing work (e.g., Lv et al., 2021; Park et al., 2023) by demonstrating how failure severity moderates the emotional responses and overall effectiveness of AI agents.

Moreover, this research expands cognitive appraisal theory by

incorporating new personal and situational factors into the framework, particularly in the context of customer-AI interactions during service failures. Recent studies have explored how service failures influence customers' perceptions of AI agents from a cognitive appraisal perspective (e.g., Cui and Zhong, 2023; Lv et al., 2021). However, these studies have primarily focused on the overall outcomes of service failures, without identifying the specific factors that shape cognitive appraisals in this process. This study introduces two key contributions to this theory. First, we identify likability as a new personal factor that mediates the relationship between anthropomorphism and failure tolerance. Second, we recognize failure severity as a situational factor that moderates the emotional response triggered by anthropomorphic elements of AI agents. Specifically, the emotional conflict between positive emotions generated by anthropomorphism and negative emotions triggered by failure severity leads to a curvilinear shift in customer tolerance behavior. These findings enrich cognitive appraisal theory by integrating emotional dynamics related to both personal and situational factors in service failure contexts, offering a more nuanced understanding of customer-AI interactions.

This study extends our understanding of uncanny valley theory by moving beyond emotional reactions to explore behavioral responses. Our findings demonstrate an inverted U-shaped effect of AI agent anthropomorphism on failure tolerance behavior. Specifically, modest levels of anthropomorphism elicit the highest failure tolerance among customers, while both high levels (i.e., closely resembling humans) and low levels (i.e., appearing highly mechanical) lead to reduced tolerance. This discovery differs from prior research that focused primarily on emotional and attitudinal responses within the framework of uncanny valley theory (Akdim et al., 2023; Blut et al., 2021; Mathur and Reichling, 2016; Stein and Ohler, 2017). This study thus provides direct evidence of the impact of AI agent anthropomorphism on customer behavior, enriching the theoretical landscape of AI-customer interactions.

7.3. Practical implications

While anthropomorphism has benefits, it's critical for managers to avoid over-anthropomorphizing AI agents. Excessive human-like features may not only reduce likability (Mathur and Reichling, 2016) but also elicit discomfort or even aversion in some customers (Mori, 1970). Our findings highlight the importance of prioritizing emotional value in AI agent design to foster favorable customer responses. Especially in the case of service failures, such highly anthropomorphic features do not necessarily bring advantages. Meanwhile, from a cost perspective, highly anthropomorphic AI agents are usually accompanied by higher design and procurement costs, which adds to the financial burden of the company. Given budget constraints, companies should allocate resources to designing AI agents that prioritize likability over striving for human-like resemblance.

Additionally, service managers should assign AI agents to service tasks based on the expected failure severity. For tasks with low-severity failure risks, anthropomorphic AI agents can be highly effective in enhancing customer emotions and tolerance for minor failures. However, for high-severity service failures, the emotional benefits of anthropomorphism may be less effective. In these cases, relying on highly anthropomorphized agents could lead to customer dissatisfaction and reduced trust if the service failure is significant. Real-time monitoring of service performance and the continuous evaluation of AI agent interactions will help managers adjust and ensure that AI agents are deployed in roles with lower failure impact, maximizing the effectiveness of anthropomorphism while minimizing risks.

8. Limitations and future directions

There are several limitations in this study. This study models a quadratic, inverted U-shaped effect of AI agent anthropomorphism on failure tolerance, focusing on a single turning point. However, the uncanny valley theory implies a cubic curve, where beyond a certain level of near-perfect anthropomorphism, individuals may exhibit a renewed positive response. Our research did not include this final phase, as current AI agents lack the technological capability to achieve flawless human resemblance. As advancements in AI make perfectly human-like digital entities feasible, future research should explore their impact on failure tolerance to fully understand the potential of anthropomorphism.

Second, this study uses a scenario-based experiment to explore the causal relationships among variables, and while this approach helps to control variables and measure them accurately, it inevitably ignores the multiple interfering factors that may be present in customer-AI interactions in real-life scenarios. Therefore, we suggest that future studies adopt the field study approach, which sets the study in a real and complex environment, to better examine the relationships between variables.

Finally, this study only focuses on hotel settings, yet AI agents are increasingly used across various sectors, where service failures also occur. Future research should extend this work to other contexts, such as restaurants or banks, to understand the effect of anthropomorphism on failure tolerance across different environments. Additionally, while we examined service failure severity as a boundary condition, other factors, such as task types, customer demographics, and market segment needs, may also influence this relationship. Exploring these conditions could deepen insights into how anthropomorphism affects failure tolerance in diverse service scenarios. Therefore, future research can further identify these potential boundary conditions.

CRediT authorship contribution statement

Ruoran Liu: Writing – review & editing, Writing – original draft, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Jin-Xing Hao:** Writing – review & editing, Validation, Supervision, Resources, Funding acquisition, Conceptualization. **Yan Yu:** Writing – review & editing, Supervision, Resources, Project administration. **Wei Shan:** Writing – review & editing, Resources, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jretconser.2025.104299.

Data availability

Data will be made available on request.

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